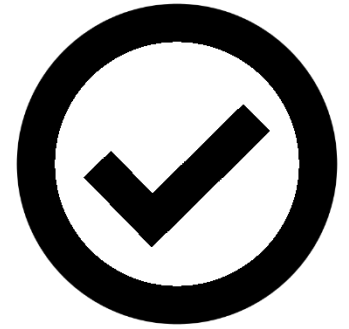




**BUSINESS
PROFESSIONALS**
of **AMERICA**
Giving Purpose to Potential



FUNDAMENTAL SPREADSHEET APPLICATIONS

(230)

STATE 2021

Production:

Job 1: Create Spreadsheet with Formulas	_____ (175 points)
Job 2: Format Spreadsheet	_____ (135 points)
Job 3: Create Pivot Table and Pivot Table Chart	_____ (50 points)
TOTAL POINTS	_____ (360 points)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

GENERAL INSTRUCTIONS:

1. Put your contestant number and printout number in the right section of the footer on each printout unless instructed otherwise. Your name or initials should *not* appear on any work you submit.
2. If you finish before the end of the testing time, notify the proctor. Time may be a factor in determining the winner in the event of a tie.
3. When turning in your contest, the jobs should be arranged in printout order.

SCENARIO:

You are a student aide in the front office of BPA HS. Your job today is to create a spreadsheet for one of the teachers that contains data on hall pass usage for her classes for the month of November.

GRADER: Points are all or none unless otherwise noted.

Job 1 (Worksheet with Formulas)	Points Possible	Points Earned
Contestant # and Job # in right section of footer	10	
11 pt. Times New Roman font used for data	10	
Set to Landscape Orientation	5	
Spreadsheet information entered	10	
Formula entered in cell G3 that calculates the total time each student is out of class <i>(Answers will not be in time format – that formatting takes place in Job 2).</i>	10	
Formula copied down to row 20	10	
Updated spreadsheet with the December information (rows 21-24)	10	
Updated the formulas to include the December data	10	
The following information was entered in the cells listed: • A26: How many passes to: • A27: Bathroom • A28: Office • A29: Nurse	20	
Function entered in Cell B27 that counts the number of bathroom visits	10	
Function entered in Cell B28 that counts the number of office visits	10	
Function entered in Cell B29 that counts the number of nurse visits	10	
In cell D27 entered the following information: “longest time out of class for one student”	5	
In cell G27, entered a function that finds the longest time a student was out of class	10	
Renamed the worksheet Mr. Norris	5	
Centered vertically and horizontally on one page	10	
Printout is in landscape view with the worksheet centered on the page	10	
Printout of formulas in landscape view with the worksheet centered on one page	10	
Subtotal	175	

Job 2 (Formatted Worksheets)	Points Possible	Points Earned
Contestant # and Job # in right section of footer	10	
Printed in landscape, centered vertically and horizontally on one sheet	10	
Formatted A1 to merge center across the data	10	
Increased the font size of the title to 20	5	
Placed a thick bottom border under the title	5	
Formatted G3:B27 to "time" with XX-XX-XX format	10	
Bolded the column titles (Date, In, Location, Etc.)	10	
G3:G24 use conditional formatting as follows: - Icon Set 3 arrows (gray) - Up arrow: >0:15:00 - Middle arrow: 0:10:00 <i>(After typing in the information, it will convert the time to a number)</i>	15	
Created a second conditional formatting on the same cell locations to highlight in yellow for any times greater than 0:15:00 minutes.	10	
Made a copy of your worksheet and renamed the new worksheet: Mr. Norris subtotal	5	
On the new worksheet, do a custom sort on first name	5	
Subtotal the data, change in first name, function count and subtotal on the total only	15	
Updated the function in cell G34 to include the total time cells for students, not the count cells	10	
Formatted this page to only print on one sheet at 93% normal size	10	
Printout of the two formatted spreadsheets centered horizontally and vertically on one page	5	
Subtotal	135	

Job 3 (Pivot Table and Pivot Chart)	Points Possible	Points Earned
Contestant # and Job # in right section of footer	10	
Based on Mr. Norris spreadsheet, a Pivot Table and Pivot Chart were created with the following fields: • Location: Legend series • First Name: Axis category • Total: Values (minutes and hours off)	15	
Placed in new worksheet; renamed Pivot Table and Chart	10	
Chart moved so the top left corner is in Cell B13	5	
Centered vertically and horizontally on one page	10	
Subtotal	50	
TOTAL POINTS	360	

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Job 1 Unformatted

BPA HS - Hall Pass Summary

Date	In Location	First Name	Last Name	Out Time	In Time	Total
2019-11-06	Bathroom	Asia	Williams	11:00:13	11:07:50	0.00528935
2019-11-06	Bathroom	Ned	Jones	09:26:34	09:37:59	0.00792824
2019-11-07	Bathroom	Kammi	Youngstown	09:32:20	09:44:21	0.00834491
2019-11-08	Nurse	Kammi	Youngstown	09:15:50	09:22:46	0.00481481
2019-11-11	Bathroom	John	Clark	09:20:41	09:27:31	0.00474537
2019-11-12	Nurse	John	Clark	09:21:02	09:24:08	0.00215278
2019-11-12	Nurse	Ned	Jones	09:36:48	09:40:37	0.00265046
2019-11-13	Bathroom	Grace	Smith	09:33:22	09:40:50	0.00518519
2019-11-13	Bathroom	Kammi	Youngstown	09:09:49	09:17:52	0.00559028
2019-11-14	Nurse	Asia	Williams	10:40:07	10:52:27	0.00856481
2019-11-14	Bathroom	John	Clark	09:22:12	09:28:03	0.0040625
2019-11-15	Bathroom	John	Clark	09:01:16	09:10:15	0.00623843
2019-11-18	Office	Asia	Williams	11:12:10	11:22:12	0.00696759
2019-11-19	Bathroom	John	Clark	09:26:07	09:32:55	0.00472222
2019-11-20	Office	Kammi	Youngstown	09:22:55	09:33:04	0.00704861
2019-11-21	Bathroom	Brandon	MacKenzie	01:14:54	01:20:06	0.00361111
2019-11-21	Bathroom	Ned	Jones	09:10:34	09:16:19	0.00399306
2019-11-22	Office	Kammi	Youngstown	08:39:31	09:03:11	0.01643519
2019-12-02	Bathroom	Ned	Jones	09:36:52	09:44:50	0.00553241
2019-12-03	Bathroom	Brandon	MacKenzie	01:32:08	01:39:15	0.00494213
2019-12-03	Office	John	Clark	09:20:00	09:23:30	0.00243056
2019-12-03	Nurse	Ned	Jones	09:01:52	09:19:58	0.01256944

How many passes to:

Bathroom	13	Longest time out of class for one student	0.0164352
Office	4		
Nurse	5		

FUNDAMENTAL SPREADSHEET APPLICATIONS

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Job 1 Formulas

BPA HS - Hall Pass Summary

Date	In Location	First Name	Last Name	Out Time	In Time	Total
2019-11-06	Bathroom	Asia	Williams	11:00:13	11:07:50	=F3-E3
2019-11-06	Bathroom	Ned	Jones	09:26:34	09:37:59	=F4-E4
2019-11-07	Bathroom	Kammi	Youngstown	09:32:20	09:44:21	=F5-E5
2019-11-08	Nurse	Kammi	Youngstown	09:15:50	09:22:46	=F6-E6
2019-11-11	Bathroom	John	Clark	09:20:41	09:27:31	=F7-E7
2019-11-12	Nurse	John	Clark	09:21:02	09:24:08	=F8-E8
2019-11-12	Nurse	Ned	Jones	09:36:48	09:40:37	=F9-E9
2019-11-13	Bathroom	Grace	Smith	09:33:22	09:40:50	=F10-E10
2019-11-13	Bathroom	Kammi	Youngstown	09:09:49	09:17:52	=F11-E11
2019-11-14	Nurse	Asia	Williams	10:40:07	10:52:27	=F12-E12
2019-11-14	Bathroom	John	Clark	09:22:12	09:28:03	=F13-E13
2019-11-15	Bathroom	John	Clark	09:01:16	09:10:15	=F14-E14
2019-11-18	Office	Asia	Williams	11:12:10	11:22:12	=F15-E15
2019-11-19	Bathroom	John	Clark	09:26:07	09:32:55	=F16-E16
2019-11-20	Office	Kammi	Youngstown	09:22:55	09:33:04	=F17-E17
2019-11-21	Bathroom	Brandon	MacKenzie	01:14:54	01:20:06	=F18-E18
2019-11-21	Bathroom	Ned	Jones	09:10:34	09:16:19	=F19-E19
2019-11-22	Office	Kammi	Youngstown	08:39:31	09:03:11	=F20-E20
2019-12-02	Bathroom	Ned	Jones	09:36:52	09:44:50	=F21-E21
2019-12-03	Bathroom	Brandon	MacKenzie	01:32:08	01:39:15	=F22-E22
2019-12-03	Office	John	Clark	09:20:00	09:23:30	=F23-E23
2019-12-03	Nurse	Ned	Jones	09:01:52	09:19:58	=F24-E24

How many passes to:

Bathroom =COUNTIF(B3:B24,"Bathroom")

Office =COUNTIF(B3:B24,"Office")

Nurse =COUNTIF(B3:B24,"Nurse")

Longest time out of class for one student

=MAX(G3:G24)

Job 2 Formatted with Subtotals**BPA HS - Hall Pass Summary**

Date	In Location	First Name	Last Name	Out Time	In Time	Total
2019-11-06	Bathroom	Asia	Williams	11:00:13	11:07:50	↓ 0:07:37
2019-11-06	Bathroom	Ned	Jones	09:26:34	09:37:59	→ 0:11:25
2019-11-07	Bathroom	Kammi	Youngstown	09:32:20	09:44:21	→ 0:12:01
2019-11-08	Nurse	Kammi	Youngstown	09:15:50	09:22:46	↓ 0:06:56
2019-11-11	Bathroom	John	Clark	09:20:41	09:27:31	↓ 0:06:50
2019-11-12	Nurse	John	Clark	09:21:02	09:24:08	↓ 0:03:06
2019-11-12	Nurse	Ned	Jones	09:36:48	09:40:37	↓ 0:03:49
2019-11-13	Bathroom	Grace	Smith	09:33:22	09:40:50	↓ 0:07:28
2019-11-13	Bathroom	Kammi	Youngstown	09:09:49	09:17:52	↓ 0:08:03
2019-11-14	Nurse	Asia	Williams	10:40:07	10:52:27	→ 0:12:20
2019-11-14	Bathroom	John	Clark	09:22:12	09:28:03	↓ 0:05:51
2019-11-15	Bathroom	John	Clark	09:01:16	09:10:15	↓ 0:08:59
2019-11-18	Office	Asia	Williams	11:12:10	11:22:12	→ 0:10:02
2019-11-19	Bathroom	John	Clark	09:26:07	09:32:55	↓ 0:06:48
2019-11-20	Office	Kammi	Youngstown	09:22:55	09:33:04	→ 0:10:09
2019-11-21	Bathroom	Brandon	MacKenzie	01:14:54	01:20:06	↓ 0:05:12
2019-11-21	Bathroom	Ned	Jones	09:10:34	09:16:19	↓ 0:05:45
2019-11-22	Office	Kammi	Youngstown	08:39:31	09:03:11	↑ 0:23:40
2019-12-02	Bathroom	Ned	Jones	09:36:52	09:44:50	↓ 0:07:58
2019-12-03	Bathroom	Brandon	MacKenzie	01:32:08	01:39:15	↓ 0:07:07
2019-12-03	Office	John	Clark	09:20:00	09:23:30	↓ 0:03:30
2019-12-03	Nurse	Ned	Jones	09:01:52	09:19:58	↑ 0:18:06

How many passes to:

Bathroom

13

Office

4

Nurse

5

Longest time out of class for one student

0:23:40

Job 2 Formatted**BPA HS - Hall Pass Summary**

Date	In Location	First Name	Last Name	Out Time	In Time	Total
2019-11-06	Bathroom	Asia	Williams	11:00:13	11:07:50	↓ 0:07:37
2019-11-14	Nurse	Asia	Williams	10:40:07	10:52:27	→ 0:12:20
2019-11-18	Office	Asia	Williams	11:12:10	11:22:12	→ 0:10:02
Asia Count						3
2019-11-21	Bathroom	Brandon	MacKenzie	01:14:54	01:20:06	↓ 0:05:12
2019-12-03	Bathroom	Brandon	MacKenzie	01:32:08	01:39:15	↓ 0:07:07
Brandon Count						2
2019-11-13	Bathroom	Grace	Smith	09:33:22	09:40:50	↓ 0:07:28
Grace Count						1
2019-11-11	Bathroom	John	Clark	09:20:41	09:27:31	↓ 0:06:50
2019-11-12	Nurse	John	Clark	09:21:02	09:24:08	↓ 0:03:06
2019-11-14	Bathroom	John	Clark	09:22:12	09:28:03	↓ 0:05:51
2019-11-15	Bathroom	John	Clark	09:01:16	09:10:15	↓ 0:08:59
2019-11-19	Bathroom	John	Clark	09:26:07	09:32:55	↓ 0:06:48
2019-12-03	Office	John	Clark	09:20:00	09:23:30	↓ 0:03:30
John Count						6
2019-11-07	Bathroom	Kammi	Youngstown	09:32:20	09:44:21	→ 0:12:01
2019-11-08	Nurse	Kammi	Youngstown	09:15:50	09:22:46	↓ 0:06:56
2019-11-13	Bathroom	Kammi	Youngstown	09:09:49	09:17:52	↓ 0:08:03
2019-11-20	Office	Kammi	Youngstown	09:22:55	09:33:04	→ 0:10:09
2019-11-22	Office	Kammi	Youngstown	08:39:31	09:03:11	↑ 0:23:40
Kammi Count						5
2019-11-06	Bathroom	Ned	Jones	09:26:34	09:37:59	→ 0:11:25
2019-11-12	Nurse	Ned	Jones	09:36:48	09:40:37	↓ 0:03:49
2019-11-21	Bathroom	Ned	Jones	09:10:34	09:16:19	↓ 0:05:45
2019-12-02	Bathroom	Ned	Jones	09:36:52	09:44:50	↓ 0:07:58
2019-12-03	Nurse	Ned	Jones	09:01:52	09:19:58	↑ 0:18:06
Ned Count						5
Grand Count						22

How many passes to:

Bathroom	13	Longest time out of class for one student	0:23:40
Office	4		
Nurse	5		

Job 3 Pivot Table and Pivot Chart

Count of Total	Column Labels			
Row Labels	Bathroom	Nurse	Office	Grand Total
Asia	1	1	1	3
Brandon	2			2
Grace	1			1
John	4	1	1	6
Kammi	2	1	2	5
Ned	3	2		5
Grand Total	13	5	4	22

